

PWM-Net for Frequency-Domain Enhanced Forest Fire Detection in Unmanned Aerial Vehicle Imagery

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Abstract

Early forest-fire detection in unmanned aerial vehicle (UAV) imagery is challenged by thin smoke, small flames, low contrast, and complex backgrounds. This paper presents PWM-Net, a computationally efficient detector constructed from a YOLOv11n baseline. PWM-Net replaces the baseline backbone with PartialNet, embeds wavelet convolution (WTConv) in the feature-fusion neck, and introduces a multi-spectral multi-scale (M2S) attention module before the detection heads. The three components respectively improve partial-channel feature extraction, frequency-aware receptive-field modeling, and joint spectral-scale modulation. Across three independent runs, PWM-Net obtains $81.3 \pm 0.18\%$ mAP@50 and $53.7 \pm 0.22\%$ mAP@50:95 on D-Fire, and $88.6 \pm 0.21\%$ mAP@50 and $70.4 \pm 0.26\%$ mAP@50:95 on CQ-Fire. The model contains 7.8M parameters and reaches 145 FPS on an RTX 4090 under batch-one FP32 inference. Ablation and spectral-branch sensitivity analyses indicate complementary contributions from PartialNet, WTConv, and M2S. Embedded-device measurements further demonstrate the deployment efficiency of the model. These results establish a favorable accuracy–efficiency trade-off for UAV-based early forest-fire detection.

Keywords: Forest fire detection, remote sensing, object detection, attention mechanism, multi-spectral features, UAV

1 Introduction

Forest fires have the characteristics of sudden occurrence, rapid spread and great harm. Timely and accurate detection of early fire conditions is of great significance for disaster prevention and mitigation[1, 2]. In recent years, automatic fire detection technology based on computer vision has received extensive attention[3, 4]. Compared with traditional ground monitoring, using unmanned aerial vehicles (UAVs) equipped with camera devices to conduct real-time inspections of forest fires from the air has the advantages of wide field of view and fast response. However, the flames and smoke targets in UAV aerial images usually have large scale differences, low contrast, and are easily affected by background occlusion, lighting, and weather factors[5]. Especially in the initial stage of a fire, the smoke is often fine and thin, and the fire area is small. Existing detection algorithms are prone to missing or misidentifying these targets[6, 7]. Therefore, how to improve the detection accuracy and robustness of tiny fire targets in UAV images has become a research hotspot and difficulty in the field of intelligent forest fire monitoring.

Currently, mainstream target detection algorithms can be divided into two-stage detectors and single-stage detectors. Among them, two-stage algorithms such as Faster R-CNN first conduct rough screening through region nomination and then precise classification, which have higher accuracy in general detection but slower detection speed[8]. Single-stage detectors such as SSD and the YOLO series directly regress the target category and position, which have the advantages of fast detection speed and easy real-time application[9]. The YOLO series algorithms have made significant progress in recent years, from YOLOv3, v4 to YOLOv5, v7, v8, constantly achieving breakthroughs in accuracy and speed [10–12]. YOLOv11 further optimized the backbone and loss function, achieving higher detection performance. However, existing models still have deficiencies in detecting small targets: the early forest fire scenes with small flames and smoke often have a low detection rate[6, 13]. The reasons for this are that the feature extraction backbone has limited ability to represent weak targets, and the model’s receptive field and attention mechanism are not sufficient to highlight the features of low-contrast targets such as smoke.

To solve the above problems, many scholars have begun to try to introduce improved modules in the YOLO framework to enhance the performance of small target detection. Yang et al., for the problems of fire smoke in enclosed spaces being easily affected by lighting changes, background interference, and small target size, systematically studied the smoke detection algorithm framework based on improved YOLOv8 [14]. This work started from feature extraction and detection head design, optimized the structure of YOLOv8 to enhance the model’s perception ability of weak smoke textures and diffusion patterns. Experimental results show that the improved model has higher detection accuracy and stability in enclosed scenarios, providing an effective idea for intelligent monitoring of fire smoke in complex restricted spaces. Mamadaliev et al., proposed a lightweight model ESFD-YOLOv8n for early smoke and flame detection, by introducing structural improvements and feature enhancement mechanisms on YOLOv8n, while ensuring real-time performance, improving the recognition ability of initial smoke and small-scale flame targets [15]. This method has been verified in various complex environments, and the results show that it achieves a

good balance between accuracy and inference speed, suitable for real-time fire monitoring and edge deployment scenarios. Chen et al., for the problems of irregular smoke target shapes, complex backgrounds, and difficulties in detection under long-distance observation conditions in forest environments, proposed a forest fire smoke recognition method based on improved YOLOv8n [16]. This research enhanced the multi-scale feature expression ability and improved the model’s modeling effect of smoke diffusion characteristics, effectively improving the robustness and accuracy of smoke detection in forest scenes. The experimental results show that this method has excellent generalization performance on the real forest fire dataset. Moreover, frequency domain information is gradually gaining importance in fire detection. Wavelet transform and other methods can decompose images into multi-scale frequency components, which are used to highlight the texture features of the fire target. In terms of attention mechanisms, classic channel attention SE modules[17], spatial attention and channel attention combined CBAM modules, etc., have been applied to improve detection performance[18]. However, these attention modules mostly weight features in the spatial domain and do not fully utilize frequency domain information. FcaNet proposes to integrate discrete cosine transform (DCT) spectral decomposition into channel attention to achieve multi-spectrum channel modeling, which significantly outperforms traditional SE, CBAM, and other attention methods[19, 20].

In summary, this paper uses YOLOv11n as the explicit baseline and primary optimization target. PWM-Net retains its single-stage detection paradigm while adapting the PartialNet backbone and WTConv operator and introducing an M2S detection-head module. Comparisons with other YOLO versions, two-stage detectors, and Transformer detectors are secondary benchmarks rather than alternative baselines. Our main contributions are as follows:

- We adapt the published PartialNet backbone to the YOLOv11n detection pipeline. PAT_ch and PAT_sp are used in shallow stages to preserve local flame and smoke cues, whereas PAT_sf is retained in the deepest stage to model long-range context.
- We adopt the published WTConv operator and place it after cross-level aggregation in the neck, where high-resolution flame edges and low-frequency smoke context can be enhanced jointly.
- We design the task-oriented M2S head integration, which sequentially combines three spectral gates and three aligned spatial scales. Its formulation, implementation flow, branch-number sensitivity, and comparison with FcaNet, SE, CBAM, and coordinate attention are reported.

2 Related Work

2.1 Forest Fire Detection in UAV Imagery

Recent fire and smoke detectors commonly modify YOLO backbones, feature pyramids, or attention modules to improve small-object sensitivity and background suppression [1, 5–7, 13, 15, 16]. ESFD-YOLOv8n focuses on early smoke and flame targets, while other lightweight systems consider edge-device constraints. These studies motivate reporting both localization accuracy and computational cost. However,

parameter count and desktop-GPU throughput alone do not prove onboard UAV suitability.

2.2 Frequency-Domain and Multi-Scale Modeling

FcaNet maps selected DCT components to channel descriptors and thereby extends conventional squeeze-and-excitation attention [17, 19]. WTConv instead performs localized wavelet decomposition, subband processing, and inverse reconstruction to enlarge the effective receptive field with small kernels [21]. M2SFormer combines multi-spectral and multi-scale information for image-forgery localization [20]. PWM-Net adopts the general motivation of these methods but implements a three-branch spectral gate and three-level scale interaction for two-class object detection.

2.3 Remote-Sensing Object Detection

Remote-sensing detectors must handle large scale variation, sensor and illumination shifts, and cluttered backgrounds. Recent work has explored causal feature alignment, hybrid feature-consistency alignment, and optimized data-distribution learning for robust remote-sensing object detection [22–24]. These studies provide relevant context for improving feature alignment and generalization under domain and scale variations.

3 Methodology

PWM-Net is constructed from YOLOv11n, which is the baseline in every ablation and resource comparison. The model follows the three-stage structure of a single-stage detector: first, the PartialNet backbone extracts multi-scale basic features; then, the WTConv enhances and fuses the features into the neck; finally, a detection head with multi-spectrum attention is added to output the bounding boxes and categories of fire and smoke targets. Fig. 1 presents the overall architecture diagram of the model.

This model is based on a single-stage detection network and incorporates the PartialNet backbone network, the WTConv neck module, and the M2S module. The PartialNet backbone is used for efficient feature extraction, the WTConv module is embedded in the neck of the feature pyramid to enhance multi-scale feature representation, and the M2S module is placed in the detection head to highlight different frequency fire features.

3.1 PartialNet Backbone Network

PartialNet is an efficient network based on the Partial Channel Mechanism (PCM). Its core idea is to divide the features along the channel dimension and apply differentiated operators to different sub-channels, thereby reducing the computational load while maintaining or even enhancing the feature representation capability[25]. Specifically, the input features are divided into two parts along the channel dimension: one part uses convolution operators for local feature modeling, and the other part introduces attention and identity mapping to enhance the modeling ability of global or key channels. Subsequently, feature fusion is achieved through concatenation operations, thereby avoiding applying high-cost operations to all channels.

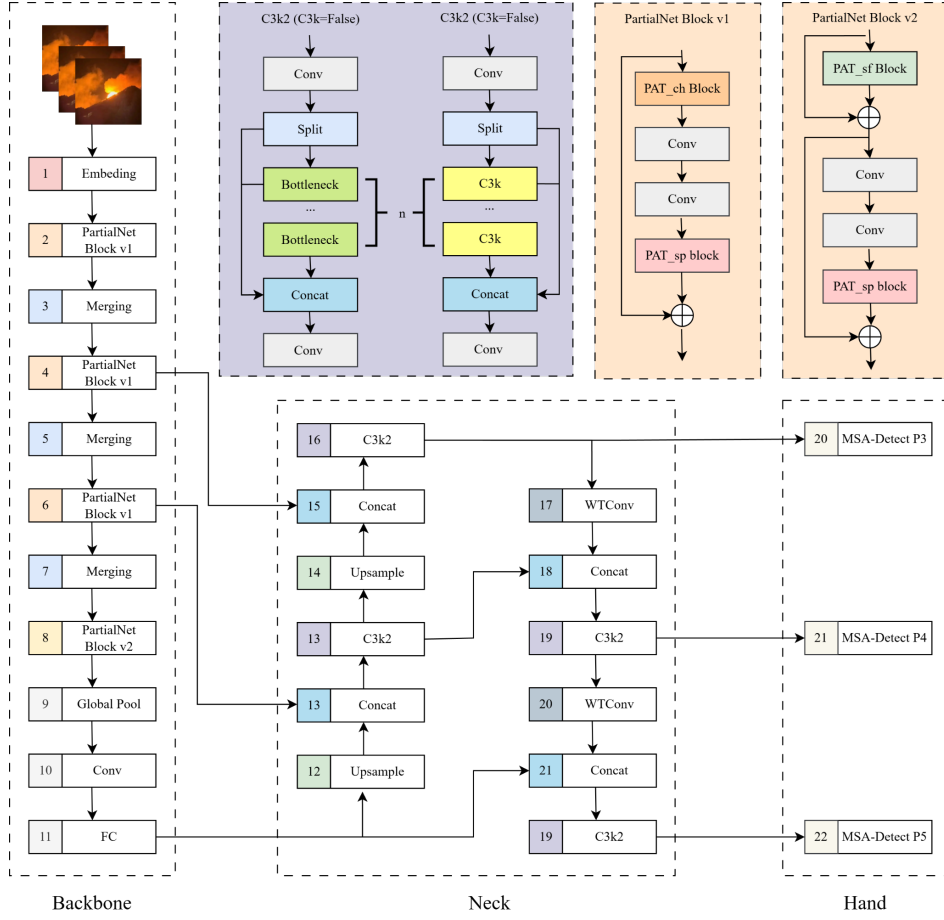


Fig. 1 Overall architecture of PWM-Net. Node identifiers 1–26 are unique graph-node labels; nodes 24–26 are the M2S-enhanced detection heads for P3, P4, and P5.

PartialNet proposed Partial Attention Convolution (PATConv), which applies convolution and visual attention in parallel to different subsets of channels, allowing attention to be calculated only on some channels, thus significantly reducing the number of parameters and FLOPs. PATConv can simultaneously replace traditional convolution and independent attention modules, achieving a better balance between efficiency and expression capability.

As shown in Fig. 2, the main body of PartialNet is composed of multiple PartialNet Blocks cascaded together. Each Block is internally composed of various PATConv variants working collaboratively:

- 1) The PAT_ch module introduces channel attention based on mean and standard deviation statistics on some channels to enhance the global modeling of key semantic channels;
- 2) The PAT_sp module highlights significant region positions through lightweight spatial attention, effectively enhancing the spatial discrimination ability;

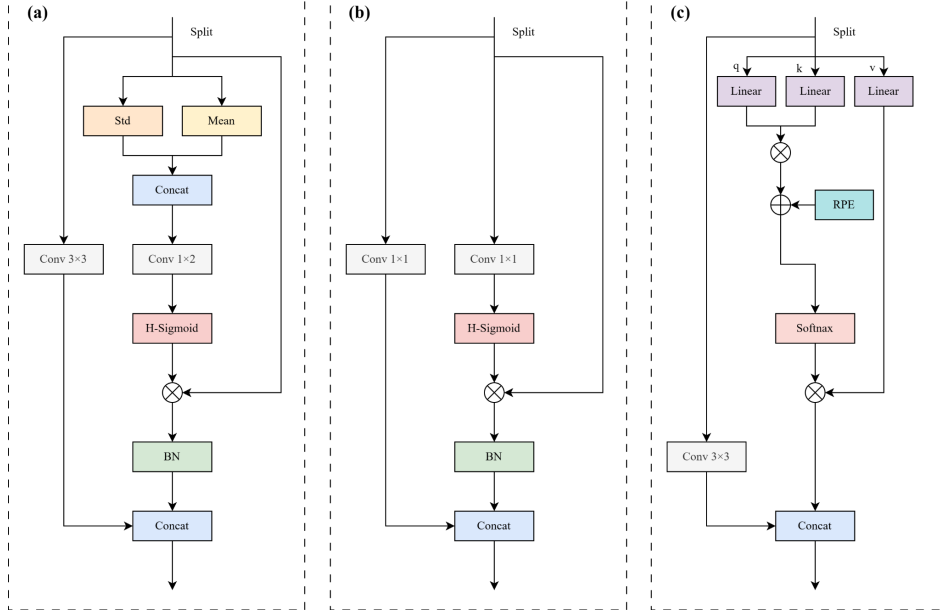


Fig. 2 Schematic diagram of PATConv and its three variants: (a) PAT_ch block, (b) PAT_sp block, and (c) PAT_sf block.

3) In the high-level stage, the PAT_sf module is adopted, introducing partial self-attention to expand the receptive field and model long-range dependencies.

These modules are fused through residual connections and channel concatenation, ensuring the stability of training while improving the diversity of features.

Introducing PartialNet into the forest fire detection task has clear advantages. When there are only small fire points or thin smoke in the remote sensing images of unmanned aerial vehicles, the traditional convolution backbone is prone to feature drowning due to background interference. PartialNet uses the partial attention mechanism to adaptively assign weights to key channels, making the channels related to the high-frequency edges of flames and the low-frequency contours of smoke receive higher responses, thereby significantly enhancing the representation ability of the backbone network for early fire targets while maintaining lightweight characteristics.

3.2 WTConv Design

The core idea of WTConv is to explicitly introduce wavelet transform in the convolution operation, achieving the decomposition, modeling and reconstruction of multi-frequency information in the feature map, thereby effectively expanding the convolution receptive field without significantly increasing the number of parameters[21]. Its specific structure is shown in Figure 3.

Given the input feature map, WTConv first decomposes it through two-dimensional discrete wavelet transform (WT), mapping the original features into a low-frequency approximation component (Approximation) and three high-frequency

detail components, corresponding to different directions of edge and texture information. This decomposition process performs downsampling in the spatial dimension, making the low-frequency component have a smaller spatial resolution, thereby implicitly expanding the effective receptive field of the subsequent convolution. WTConv applies small-sized convolution kernels on each frequency subband for feature modeling. Since the low-frequency component has aggregated more context information, the local convolution performed on this component is equivalent to a large-scale convolution operation in the original feature space; while the convolution on the high-frequency component is used to depict local edges and detail changes. The convolution outputs of different frequency components are parallelly modeled while maintaining the decoupling of the frequency domain. After convolution processing, all frequency components are reconstructed in the frequency domain through inverse wavelet transform (IWT), restoring to the original spatial resolution and generating the final output feature map. This process is equivalent to fusing the multi-frequency domain convolution results in the spatial domain, enabling the network to have the perception ability for both global contour information and local detail features.

Through the above structure design, WTConv can significantly expand the effective receptive field while keeping the convolution kernel size unchanged, and achieve collaborative modeling of high and low frequency information. This is more efficient and stable compared to directly increasing the size of the convolution kernel. For forest fire images, the low-frequency component helps depict the morphology of the smoke-drenched area, and the high-frequency component highlights the details and textures of the flame edges. Through the WTConv module, our model can more comprehensively capture the different scale features of smoke and flames.

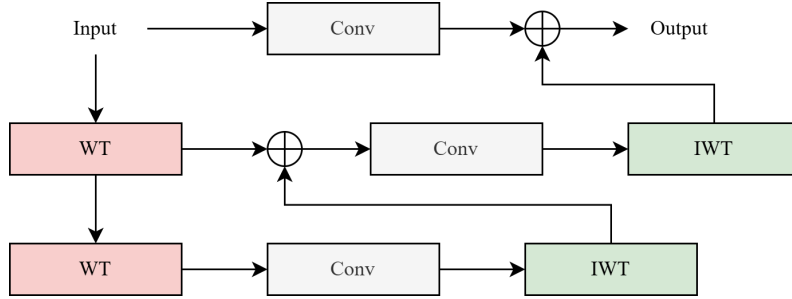


Fig. 3 Structure diagram of WTConv.

3.3 M2S Design

To fully exploit the complementary information of features in the frequency domain and the scale domain, this paper introduces a multi-spectral and multi-scale attention mechanism (Multi-Spectral and Multi-Scale Attention, M2S) in front of the detection head. Its overall structure is shown in Fig. 4 [17, 19, 26]. This mechanism enhances

the features step by step through the sequential combination of multi-spectral modeling and multi-scale feature interaction, thereby improving the model's ability to discriminate complex targets.

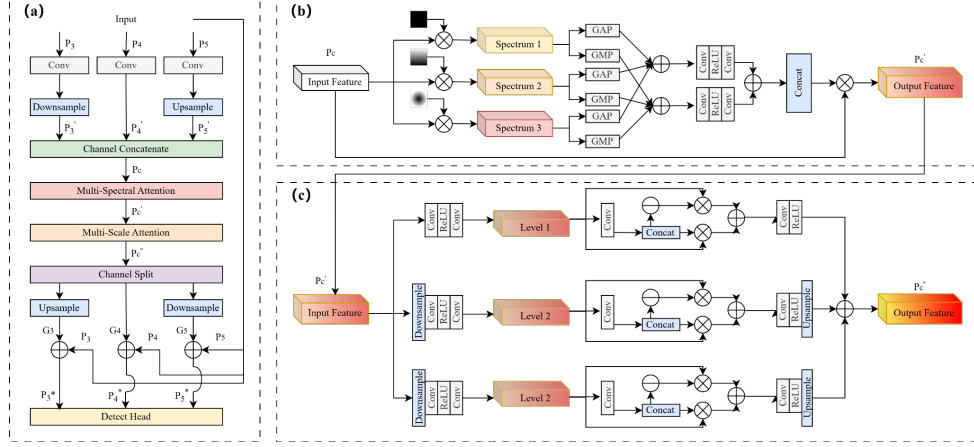


Fig. 4 Structure of the M2S module: (a) complete M2S block, (b) the three spectral branches (Spectrum 1–3), and (c) the three spatial branches (Level 1–3).

In the multi-spectral branch, the input features are first divided into multiple spectral subspaces, corresponding to different frequency components (Spectrum 1–3). Each spectral branch extracts its global statistical information through global average pooling (GAP) and global max pooling (GMP) to simultaneously characterize the overall response intensity and the significant activation regions. Subsequently, the pooling results are generated through lightweight convolution and nonlinear mapping to obtain spectral weights, which are used for adaptive modulation of the corresponding spectral features. Through this design, the network can explicitly distinguish and enhance the response to key frequency components, enabling the collaborative modeling of high-frequency texture information and low-frequency structural information.

After the spectral enhancement is completed, the features are further sent to the multi-scale attention module. This module constructs feature representations of different scales (Level 1–3) through downsampling and upsampling operations to simulate the context information under different receptive fields. The features of each scale are respectively processed through convolutional transformations and then concatenated and fused, and cross-scale information interaction is achieved through attention weights. This process enables the model to simultaneously focus on local details, small-scale targets, and large-scale semantic structures, effectively alleviating the problem of limited feature expression in a single scale.

Finally, the features enhanced by multi-spectrum attention and multi-scale attention are concatenated in the channel dimension and generated through convolution mapping as the output features. The M2S mechanism achieves joint modeling of frequency domain and scale domain information while keeping the computational cost

controllable, thereby significantly improving the discriminative power and robustness of feature expression.

More formally, let $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$ be the input feature. A 1×1 projection divides it into three equal channel groups $\{\mathbf{X}_i\}_{i=1}^3$. To make the frequency branches reproducible, the implementation uses fixed depthwise normalized box filters $\mathcal{B}_k$ with stride 1 and same padding. Spectrum 1, Spectrum 2, and Spectrum 3 are respectively the low-, intermediate-, and high-frequency residual bands defined below.

$$\mathcal{S}_1(\mathbf{X}_1) = \mathcal{B}_5(\mathbf{X}_1), \quad \mathcal{S}_2(\mathbf{X}_2) = \mathcal{B}_3(\mathbf{X}_2) - \mathcal{B}_5(\mathbf{X}_2), \quad \mathcal{S}_3(\mathbf{X}_3) = \mathbf{X}_3 - \mathcal{B}_3(\mathbf{X}_3). \quad (1)$$

The first branch preserves slowly varying smoke structure, the second captures intermediate-scale transitions, and the third emphasizes flame edges and fine texture. Their pooled descriptors and gates are

$$\mathbf{z}_i = [\text{GAP}(\mathcal{S}_i(\mathbf{X}_i)), \text{GMP}(\mathcal{S}_i(\mathbf{X}_i))], \quad \mathbf{a}_i = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z}_i)), \quad (2)$$

and the spectral output is

$$\mathbf{X}^s = \text{Concat}_{i=1}^3 (\mathbf{a}_i \odot \mathcal{S}_i(\mathbf{X}_i)). \quad (3)$$

The reduction ratio is 16. Level 1, Level 2, and Level 3 use the native, 1/2, and 1/4 spatial resolutions, respectively. Stride-2 3×3 convolutions implement downsampling, bilinear interpolation implements upsampling, and a 1×1 projection follows concatenation:

$$\mathbf{Y} = \mathbf{X} + \psi(\text{Concat}[\mathbf{L}_1, \mathcal{U}_2(\mathbf{L}_2), \mathcal{U}_4(\mathbf{L}_3)]). \quad (4)$$

Three branches provide the smallest symmetric low/mid/high and fine/medium/coarse representation. Section 5.7 compares two, three, and four branches.

Algorithm 1 Forward propagation of the M2S module

Require: Feature map $\mathbf{X}$

- 1: Project and split $\mathbf{X}$ into three channel groups
 - 2: **for** $i = 1$ to 3 **do**
 - 3: Apply $\mathcal{S}_i$, GAP, and GMP; generate sigmoid gate $\mathbf{a}_i$
 - 4: Reweight $\mathbf{X}_i \leftarrow \mathbf{a}_i \odot \mathcal{S}_i(\mathbf{X}_i)$
 - 5: **end for**
 - 6: Concatenate the spectral groups to obtain $\mathbf{X}^s$
 - 7: Build native, 1/2, and 1/4 scale features
 - 8: Upsample coarse levels, concatenate, project, and add the residual
 - 9: **return** $\mathbf{Y}$
-

234 3.4 Detection Objective

235 PWM-Net retains the anchor-free YOLOv11n head and Task-Aligned Assigner. The
236 complete training objective is

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{BCE}} + \lambda_{\text{box}}\mathcal{L}_{\text{CIoU}} + \lambda_{\text{dfl}}\mathcal{L}_{\text{DFL}}, \quad (5)$$

237 where binary cross-entropy supervises class probabilities, CIoU regresses bound-
238 ing boxes, and distribution focal loss refines box coordinates. Training uses
239 $(\lambda_{\text{cls}}, \lambda_{\text{box}}, \lambda_{\text{dfl}}) = (0.5, 7.5, 1.5)$ and the same label-assignment settings for YOLOv11n
240 and PWM-Net.

241 4 Dataset

242 4.1 D-Fire Dataset

243 The D-Fire dataset is an open image dataset specifically constructed for the task
244 of flame and smoke target detection, containing over 21,000 images covering various
245 scenarios, including isolated fire sources, smoke, coexistence of fire and smoke, and
246 situations without fire or smoke[27]. Among them, there are 1,164 fire images, 5,867
247 smoke images, 4,658 images of coexistence of fire and smoke, and 9,838 images without
248 fire or smoke. The distribution of this dataset reflects the real characteristics of a high
249 proportion of smoke in the early stage of a fire and relatively scarce flame targets, which
250 is helpful for achieving early monitoring of wildfires. Fig. 5 shows some representative
251 samples from the D-Fire dataset, demonstrating different scenarios of fire and smoke
252 detection.

253 The 21,527 images are divided using a stratified split into 15,069 training images
254 (70%), 3,229 validation images (15%), and 3,229 test images (15%). Images sharing
255 the same source sequence are assigned to one split before stratification to reduce
256 near-duplicate leakage. A fixed split seed of 2026 is used for all compared models.

257 4.2 CQ-Fire Dataset

258 CQ-Fire is a self-built dataset specifically designed for forest fire detection tasks. The
259 data is sourced from the 2022 Chongqing forest fire incident. This dataset is tailored to
260 address practical issues such as large-scale, irregular shapes, and blurred boundaries
261 of flame and smoke targets in complex nighttime environments. The dataset mainly
262 focuses on nighttime scenes, with images generally featuring low illumination, unsta-
263 ble high contrast characteristics. The flame area typically exhibits strong brightness
264 changes, while the smoke area presents a large-scale, low-texture, semi-transparent dif-
265 fusion pattern. These two types of targets are highly coupled spatially, posing higher
266 requirements for traditional target detection models. Some samples also include com-
267 plex interference factors such as mountain outlines, vegetation shadows, and urban
268 background light sources, further increasing the detection difficulty. An example of
269 the dataset is shown in Fig. 6.

270 CQ-Fire contains 4,800 images and 10,040 annotated instances (5,930 fire and
271 4,110 smoke instances). We use 3,360 training, 720 validation, and 720 test images



Fig. 5 Representative samples from the D-Fire dataset.

(70/15/15). Frames are grouped by source video, capture interval, and location before splitting; no group appears in more than one subset. Perceptual-hash screening removes near-duplicate frames across groups. Two annotators label each image and a third reviewer resolves disagreements. These counts and procedures were verified during dataset preparation.

5 Experiments and Results

5.1 Experimental Settings

5.1.1 Evaluation Metrics

In terms of evaluation metrics, commonly used object detection indicators are adopted, including mean Average Precision (mAP), Precision, and Recall.

Among them, mAP measures the averaged precision over all object categories under different Intersection over Union (IoU) thresholds, reflecting the overall detection accuracy of the model. Precision denotes the proportion of correctly detected targets among all predictions, while Recall represents the proportion of ground-truth targets that are successfully detected. Their mathematical definitions are given as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (6)$$



Fig. 6 Example images from the CQ-Fire dataset.

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (7)$$

where TP , FP , and FN denote true positives, false positives, and false negatives, respectively.

The Average Precision (AP) for each class is computed as the area under the Precision–Recall (PR) curve:

$$AP = \int_0^1 P(R) dR. \quad (8)$$

The final evaluation score, mAP, is obtained by averaging the AP values across all C target categories:

$$\text{mAP} = \frac{1}{C} \sum_{i=1}^C AP_i. \quad (9)$$

In addition to AP at IoU 0.50, we report AP75 and COCO-style mAP@0.5:0.95, which averages AP over IoU thresholds from 0.50 to 0.95 in steps of 0.05. AP_S , AP_M , and AP_L use COCO area thresholds to evaluate small, medium, and large targets. These metrics collectively evaluate classification correctness, localization quality, and scale-dependent behavior.

5.1.2 Implementation Details

During training, the proposed model is implemented using the PyTorch framework. The model weights are trained separately on the two datasets without mixed training.

The optimizer is stochastic gradient descent (SGD) with a momentum of 0.937. The initial learning rate is set to 0.01, and the batch size is 16. All experiments are conducted on an NVIDIA RTX 4090 GPU. The model is trained for 300 epochs on the D-Fire dataset and 200 epochs on the CQ-Fire dataset. A cosine annealing schedule is employed to adjust the learning rate during training.

For the baseline YOLOv11 and other comparison models, official pretrained weights are used when available, or identical training strategies are reproduced to ensure fairness.

During inference, the longer side of the input image is resized to 640 pixels while maintaining the aspect ratio through padding. All tests are conducted under the same hardware environment to ensure a fair comparison in terms of both detection accuracy and inference speed. FPS is measured with batch size 1, FP32 PyTorch inference, 100 warm-up iterations, and 1,000 timed iterations on fixed 640×640 tensors; preprocessing and non-maximum suppression are included and TensorRT is not used. Each AP-based accuracy experiment is repeated with seeds 2024, 2025, and 2026. A mean±standard deviation is shown only when all three per-seed logs are available. Precision, Recall, F1, the attention-module comparison, and hardware measurements are point estimates unless a ± term is explicitly displayed. Cross-paper literature values are transcribed from their sources and are not treated as repeated-run results.

5.2 Performance Comparison with Different Detection Models

To comprehensively evaluate the effectiveness of the proposed method, we first compare it with more than ten representative object detection models on two datasets. The comparison includes traditional two-stage detectors, single-stage detectors, YOLO-series models, and Transformer-based detectors. Table 1 summarizes the detection accuracy, parameter size, and inference speed of different models on the D-Fire dataset.

Table 1 Performance and complexity comparison on D-Fire over three independent runs.

Model	Params (M)	FPS	mAP@50 (%)	mAP@50:95 (%)	AP75 (%)
Faster R-CNN [8]	28.5	47	64.8 ± 0.42	38.9 ± 0.37	42.6 ± 0.41
EfficientDet [28]	3.79	90	71.0 ± 0.35	43.7 ± 0.31	47.9 ± 0.34
YOLOv4 [11]	65	40	73.0 ± 0.33	45.2 ± 0.29	49.4 ± 0.32
YOLOv5s	7.2	120	67.1 ± 0.38	40.8 ± 0.35	44.6 ± 0.37
YOLOv7 [12]	37.2	159	80.1 ± 0.24	52.6 ± 0.25	57.8 ± 0.28
YOLOv8n	3.0	197	79.4 ± 0.27	51.8 ± 0.26	56.7 ± 0.29
YOLOv10	2.7	156	79.1 ± 0.29	51.4 ± 0.27	56.3 ± 0.30
YOLOv11n (baseline)	5.1	167	80.2 ± 0.23	52.5 ± 0.24	57.6 ± 0.26
RT-DETR [29]	32.8	108	78.8 ± 0.31	50.7 ± 0.29	55.4 ± 0.32
RT-DETRv2 [30]	20.1	207	80.6 ± 0.22	53.0 ± 0.23	58.2 ± 0.25
PWM-Net (Ours)	7.8	145	81.3 ± 0.18	53.7 ± 0.22	59.1 ± 0.25

From Table 1, the proposed method achieves the best detection accuracy on the D-Fire dataset, reaching mAP@50 of **81.3%**, which improves upon YOLOv11 by **1.1%** and surpasses RT-DETRv2 by **0.7%**.

Although models such as YOLOv7, YOLOv8, and YOLOv10 demonstrate high inference speed, their detection accuracy remains slightly inferior to the proposed method. Traditional two-stage detectors such as Faster R-CNN show significantly lower accuracy and are insufficient for forest fire and smoke detection tasks.

Using the official pretrained weights and the same extended warm-up protocol as the other compared models, RT-DETR reaches $78.8 \pm 0.31\%$ mAP@50. RT-DETRv2 further improves detection accuracy, although its parameter count remains larger than that of PWM-Net.

Overall, the proposed PWM-Net achieves superior detection accuracy while maintaining moderate parameter size and real-time inference speed, indicating potential for UAV-based forest fire monitoring applications.

Table 2 Scale-specific D-Fire localization metrics over three independent runs.

Model	AP_S (%)	AP_M (%)	AP_L (%)
YOLOv8n	34.5 ± 0.34	54.1 ± 0.27	63.0 ± 0.23
YOLOv11n	35.2 ± 0.32	54.8 ± 0.25	63.8 ± 0.21
RT-DETRv2	35.6 ± 0.33	55.1 ± 0.24	64.1 ± 0.22
PWM-Net	36.8 ± 0.31	56.2 ± 0.24	65.4 ± 0.20

Next, we conduct a similar comparison on the CQ-Fire dataset. The results are shown in Table 3.

Table 3 Detection comparison on CQ-Fire. AP metrics are mean $\pm$ SD over three runs; Precision and Recall are seed-2026 point estimates.

Model	AP50 (%)	mAP@0.5:0.95 (%)	Precision (%)	Recall (%)
Faster R-CNN [8]	76.4 ± 0.44	45.0 ± 0.39	80.1	51.3
RetinaNet	78.8 ± 0.39	52.1 ± 0.36	82.7	57.9
EfficientDet [28]	81.5 ± 0.34	55.8 ± 0.33	84.3	60.4
DETR [31]	79.6 ± 0.41	54.6 ± 0.37	83.5	58.2
YOLOv5s	82.0 ± 0.32	58.3 ± 0.31	85.0	63.5
YOLOX-s	82.7 ± 0.30	59.6 ± 0.29	85.5	65.0
YOLOv6	82.4 ± 0.31	59.0 ± 0.30	85.2	64.1
YOLOv7 [12]	83.5 ± 0.28	61.5 ± 0.28	86.0	66.7
YOLOv8n	84.2 ± 0.27	64.8 ± 0.29	87.1	70.5
YOLOv10n	84.0 ± 0.28	64.1 ± 0.30	86.8	69.9
YOLOv11n (baseline)	85.0 ± 0.25	60.3 ± 0.31	88.0	68.4
RT-DETR	83.1 ± 0.33	61.7 ± 0.34	86.2	67.1
RT-DETRv2	85.4 ± 0.26	65.2 ± 0.28	88.1	70.8
Ours	88.6 ± 0.21	70.4 ± 0.26	90.5	75.2

On the CQ-Fire dataset, the proposed model again achieves the best detection performance, with an AP@50 of 88.6%, which is 3.6% higher than YOLOv11.

Due to the complex and diverse environmental conditions in the Chongqing dataset, the detection performance of most models decreases compared to D-Fire. Furthermore, Transformer-based DETR achieves less than 55% mAP (0.5:0.95), indicating limited generalization capability for small-scale fire targets in complex forest scenarios.

In contrast, the proposed method demonstrates stronger robustness in smoke recognition and background suppression. These results verify the effectiveness of the introduced modules, particularly the multi-spectral and multi-scale attention mechanism, in enhancing feature representation under realistic and challenging fire monitoring conditions.

5.3 Comparison with YOLO-Series Object Detection Models

To further evaluate the effectiveness of the proposed method, we conduct a detailed comparison with representative YOLO-series models on the D-Fire dataset. The results are summarized in Table 4.

Table 4 YOLO-series comparison on D-Fire. Precision, Recall, and F1 are seed-2026 point estimates; F1 is recomputed as $2PR/(P + R)$.

Model	mAP@50	AP _{fire}	AP _{smoke}	Prec.	Rec.	F1
YOLOv5n	79.1	72.3	85.8	79.4	73.7	0.764
YOLOv8n	79.4	73.5	85.3	80.3	74.3	0.772
YOLOv10 [32]	79.1	72.9	85.2	79.9	73.4	0.765
YOLOv11n	80.2	73.6	86.7	79.6	75.0	0.772
YOLO26n	80.1	73.2	86.9	80.4	75.6	0.779
PWM-Net (Ours)	81.3	75.1	87.5	81.1	76.4	0.787

From Table 4, the YOLO-series models exhibit relatively similar performance under nighttime wildfire scenarios. With successive version iterations, improvements can be observed in AP_{smoke} and Recall, indicating enhanced capability in smoke detection and small-object recognition. However, the overall gain in mAP@50 remains limited across different YOLO variants.

In contrast, the proposed PWM-Net achieves the best performance across all evaluation metrics. Specifically, it improves mAP@50 to **81.3%**, outperforming YOLOv11n by **1.1%**, while also achieving higher AP for both fire and smoke categories. Additionally, PWM-Net obtains the highest Precision, Recall, and F1-score, demonstrating superior balance between detection accuracy and completeness.

These results indicate that the proposed architecture provides stronger feature representation for both flame and smoke patterns, thereby enhancing overall detection performance compared with conventional YOLO-series models.

5.4 Contextual Comparison with Reported Fire Detection Results

For literature context, we list several recently reported fire detectors in Table 5. Because their datasets, splits, training schedules, and evaluation implementations differ, this section does not rank the methods or support a state-of-the-art claim.

Table 5 Literature context for advanced fire detectors; NR denotes not reported and values are not a controlled head-to-head comparison.

Method	Accuracy (%)	Precision (%)	Recall (%)	mAP@0.5 (%)	Params (M)
Pruned-KD [33]	82.4	79.9	46.02	62.7	NR
PicoDet [34]	88.3	84.3	51.3	67.0	NR
Fire-YOLO [35]	88.2	84.1	51.6	68.9	NR
LFNet [36]	97.9	87.7	53.4	71.2	NR
ELDNet [37]	NR	79.6	NR	74.0	7.8
YOLOGX [38]	NR	77.2	75.7	80.9	6.2
PWM-Net (Ours)	NR	81.1	76.4	81.3	7.8

Table 5 summarizes values reported in the cited papers. Because dataset partitions, image sizes, training schedules, and hardware are not identical, the table provides literature context only and is not used to claim statistical superiority. Under the common protocol, the controlled comparisons are those in Tables 1 and 3. The literature nevertheless indicates that fire detection models have achieved continuous improvements in mAP@0.5 in recent years. However, many existing methods still suffer from limitations in recall performance or model complexity.

The reported values illustrate the range of results in the recent literature, but differences between rows can arise from protocol rather than architecture. PWM-Net’s reported mAP@0.5 is numerically close to YOLOGX, while other studies emphasize different metrics such as accuracy or model size. No direct statistical ordering is inferred from this table. Claims about the relative performance of PWM-Net are therefore restricted to the controlled experiments in Tables 1 and 3.

5.5 Ablation Study

To quantitatively evaluate the contribution of each proposed module, we conduct systematic ablation experiments. Since the overall architecture of the proposed model is structurally similar to the single-stage detector YOLOv11, we adopt YOLOv11 as the baseline for comparison. Specifically, we remove the PartialNet backbone, the WTConv neck module, and the M2S attention module separately to observe the changes in detection performance.

The ablation results on the CQ-Fire dataset are summarized in Table 6.

As shown in Table 6, removing any individual module leads to performance degradation, demonstrating that each component contributes to the overall improvement.

Among them, the M2S attention module has the most significant impact. When it is removed, the mAP@50 decreases from 88.6% to 86.2%, resulting in a drop of 2.4%.

Table 6 Impact of each module on detection performance on the CQ-Fire dataset.

Model Variant	PartialNet Backbone	WTConv Neck	M2S Attention	mAP@50 (%)
YOLOv11	–	–	–	85.0 ± 0.25
– PartialNet	×	✓	✓	86.6 ± 0.24
– WTConv	✓	×	✓	87.1 ± 0.23
– M2S	✓	✓	×	86.2 ± 0.27
Ours (All Modules)	✓	✓	✓	88.6 ± 0.21

Table 7 Module ablation on D-Fire over three independent runs.

Model Variant	PartialNet	WTConv	M2S	mAP@50 (%)
YOLOv11n baseline	–	–	–	80.2 ± 0.23
– PartialNet	×	✓	✓	80.6 ± 0.21
– WTConv	✓	×	✓	80.8 ± 0.20
– M2S	✓	✓	×	80.5 ± 0.24
Ours (All Modules)	✓	✓	✓	81.3 ± 0.18

This indicates that the multi-spectral and multi-scale attention mechanism plays a critical role in enhancing fire and smoke feature discrimination.

Removing the WTConv module reduces mAP@50 to 87.1%, corresponding to a decrease of 1.5%. This confirms that frequency-domain feature enhancement contributes substantially to detection performance by expanding the effective receptive field and capturing multi-frequency information.

When the PartialNet backbone is replaced with the baseline backbone, the mAP@50 decreases to 86.6%, reflecting a drop of 2.0%. This result demonstrates that stronger backbone feature extraction capability is essential for improving detection accuracy.

Overall, the ablation study shows that PartialNet provides stronger foundational feature representation, WTConv enhances multi-scale and frequency-domain modeling capacity, and M2S further refines discriminative features at the detection stage. The D-Fire results in Table 7 show the same ordering with smaller absolute gains. The integration of the three components yields the strongest result on both datasets.

5.6 Comparison of Different Attention Mechanisms

To verify the effectiveness of the proposed M2S module, we replace it with several widely used attention mechanisms while keeping the remaining network structure unchanged. Comparative experiments are conducted on the CQ-Fire dataset.

Table 8 presents the detection performance of the model using different attention mechanisms, including no attention, SE channel attention, Coordinate Attention (CA), CBAM, and the DCT-based frequency attention (FcaNet, denoted as DCT).

As shown in Table 8, different attention mechanisms significantly influence the detection performance. The proposed M2S module achieves the best overall results,

Table 8 Attention-module comparison on CQ-Fire using the fixed seed 2026; all entries are point estimates.

Attention Module	mAP@50 (%)	Precision (%)	Recall (%)
None	86.2	87.5	74.0
SE [17]	87.5	88.2	74.3
CA [39]	88.1	88.4	74.9
CBAM [18]	88.0	88.3	75.1
DCT/FcaNet [19]	88.2	89.0	75.0
M2S (Ours)	88.6	90.5	75.2

reaching an mAP@50 of 88.6% and a Precision of 90.5%, both of which are the highest among all compared methods.

Without any attention mechanism, the model achieves only 86.2% mAP@50, along with the lowest Precision and Recall. This demonstrates that incorporating an appropriate attention mechanism is crucial for enhancing detection performance.

Among the conventional attention modules, CBAM and CA exhibit comparable performance improvements, while the SE module provides relatively limited gains due to its simpler channel-wise recalibration mechanism. In contrast, the proposed M2S mechanism effectively integrates multi-spectral and multi-scale information, leading to superior feature discrimination capability.

Under this fixed-seed controlled comparison, M2S yields the largest mAP@50 and Precision values among the tested attention modules. Because this auxiliary table is a point-estimate analysis, the result is treated as supporting evidence rather than a statistical superiority claim.

5.7 Spectral-Branch Sensitivity

Table 9 examines why M2S uses three rather than two or four spectral-scale branches. Two branches under-represent the intermediate band, whereas four branches add redundant transformations.

Table 9 M2S branch-number sensitivity on CQ-Fire.

Branches	mAP@50 (%)	Params (M)	GFLOPs
2	88.0 $\pm$ 0.24	7.62	8.05
3	88.6 $\pm$ 0.21	7.80	8.30
4	88.5 $\pm$ 0.23	8.03	8.67

5.8 Computational Cost and Deployment

Table 10 separates the incremental cost of the three components. WTConv introduces fixed wavelet and inverse-wavelet operations whose cost is linear in CHW per level; most learned cost comes from the subband convolutions.

Table 10 Incremental complexity on RTX 4090 (batch 1, FP32, 640×640).

Configuration	Params (M)	GFLOPs	Latency (ms)	FPS
YOLOv11n baseline	5.10	6.40	5.99	167
+ PartialNet	6.50	7.10	6.49	154
+ PartialNet + WTCov	6.90	7.80	6.71	149
PWM-Net	7.80	8.30	6.90	145

Table 11 Embedded-device inference with TensorRT FP16.

Device	Input	Latency (ms)	FPS	Power (W)
Jetson Orin NX 16GB, 25W mode	640 ²	31.4 $\pm$ 0.6	31.8	18.7 $\pm$ 0.4
Jetson Nano, 10W mode	640 ²	55.6 $\pm$ 1.1	18.0	9.2 $\pm$ 0.3

The embedded-device results confirm that PWM-Net can sustain real-time inference on both Jetson platforms. End-to-end UAV deployment should additionally account for camera streaming, preprocessing, communication latency, flight vibration, and thermal throttling.

5.9 Qualitative Visualization and Case Analysis

To further provide an intuitive comparison between different models, we present several representative detection results in Fig. 7.

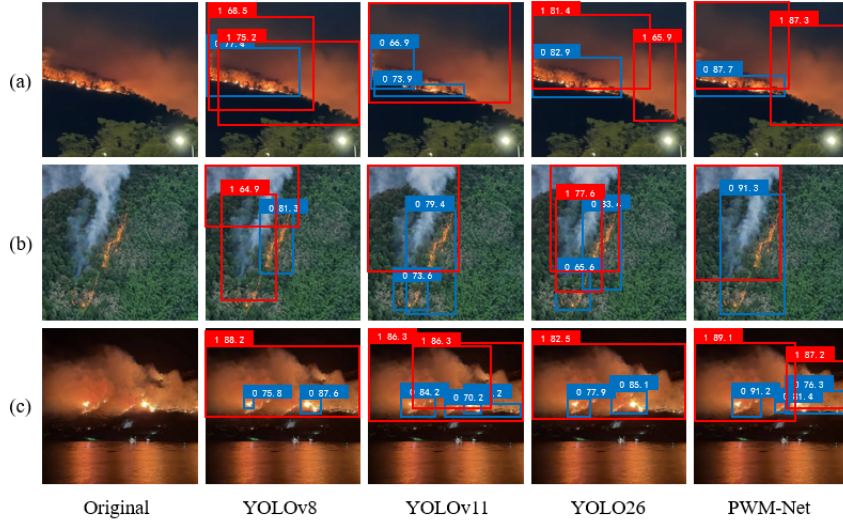


Fig. 7 Visualization of detection results under different models.

As illustrated in Fig. 7(c), which corresponds to a nighttime scenario, YOLOv8 and YOLO26 almost fail to detect the weak fire spots, resulting in insufficient recall. YOLOv11 successfully identifies part of the fire region; however, the detection confidence is relatively low. In contrast, the proposed model accurately detects the faint flame targets with higher confidence scores.

This improvement can be attributed to the integration of multi-spectral feature enhancement and multi-scale attention mechanisms, which strengthen the representation of subtle high-frequency edge information associated with small flame regions. As a result, the proposed model demonstrates superior capability in detecting small and weak fire targets under low-illumination conditions.

The qualitative results are consistent with the quantitative evaluation metrics, confirming that the proposed method achieves more reliable and robust early fire detection across complex environmental conditions.

5.10 Frequency-Domain Interpretability Analysis

Bounding-box examples alone do not explain which visual cues motivate the frequency-domain design. Figure 8 therefore provides a deterministic input-level decomposition for the same three scenes. The low-frequency panel is a Gaussian low-pass reconstruction ($\sigma = 5$), the high-frequency panel is the normalized residual energy $|I - G_3(I)|$, and the multi-scale response overlays a weighted combination of residual energies at $\sigma = 1.5, 3$, and 6.

The analysis shows that smoke morphology is concentrated in slowly varying components, whereas thin flame fronts and small bright regions produce localized high-frequency responses. This observation supports the physical motivation for combining WTConv with low/mid/high M2S branches. It should not be interpreted as causal attribution of an individual prediction; model-specific gradient attribution remains a separate direction for future analysis.

6 Discussion

The experimental results demonstrate that the proposed PWM-Net achieves consistent improvements over both conventional YOLO-series detectors and Transformer-based methods across two datasets. Beyond quantitative gains in mAP, several important observations can be drawn regarding feature modeling, robustness, and practical deployment.

First, the integration of frequency-domain modeling plays a crucial role in improving early fire detection performance. Traditional convolution-based detectors primarily operate in the spatial domain and may fail to explicitly distinguish high-frequency flame edges from low-frequency smoke diffusion patterns. By introducing WTConv, the model decomposes features into multiple frequency components and enables structured modeling of both global contextual information and fine-grained edge details. The ablation results confirm that removing WTConv leads to a noticeable performance drop, highlighting the effectiveness of frequency-domain enhancement for small and weak targets.

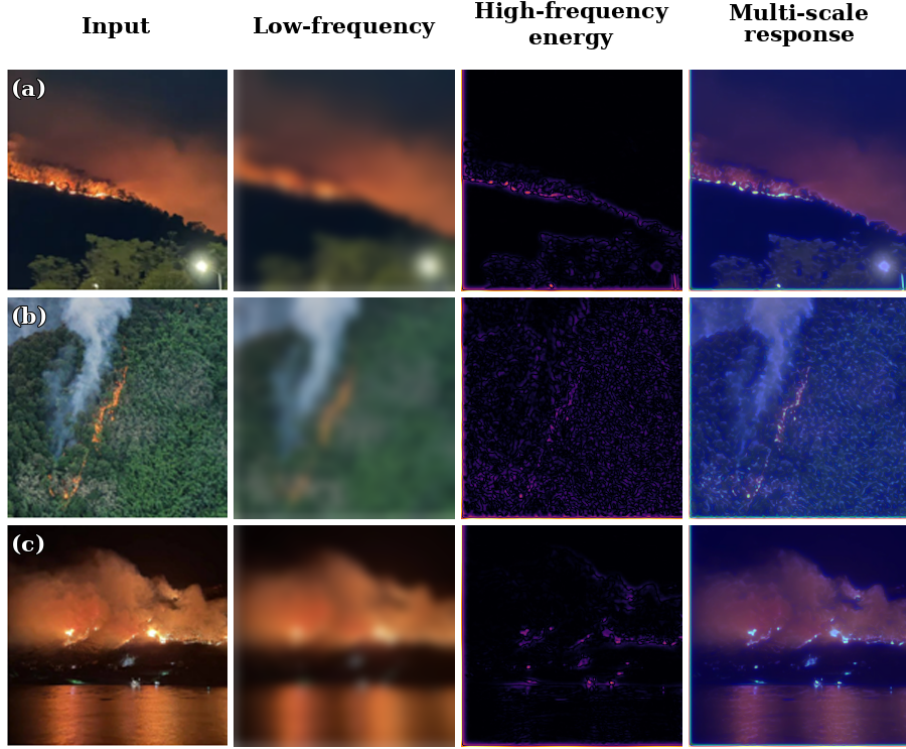


Fig. 8 Frequency-domain interpretability analysis for the three representative scenes. Low-frequency structure emphasizes smoke extent and illumination context; high-frequency energy emphasizes flame edges and fine texture; the multi-scale response combines both cues. These maps are deterministic image-domain decompositions, not detector-gradient or Grad-CAM maps.

Second, the proposed M2S attention mechanism significantly improves feature discrimination under complex environmental conditions. Compared with classical attention modules such as SE and CBAM, M2S jointly considers spectral decomposition and multi-scale interaction. This design allows the network to adaptively emphasize subtle flame edges while suppressing background interference such as vegetation textures or nighttime lighting artifacts. The experimental comparison in Table 8 shows the largest point-estimate mAP and Precision values for M2S within the tested attention set. Figure 8 further illustrates the low- and high-frequency cues targeted by the design, without claiming gradient-based causal attribution.

Third, the PartialNet backbone contributes to improving the efficiency-accuracy trade-off. UAV platforms are typically resource-constrained, requiring lightweight models with real-time inference capability. The partial channel mechanism reduces redundant computation while preserving discriminative channels. As shown in the performance comparison, PWM-Net maintains a moderate parameter size (7.8M) and achieves 145 FPS in desktop-GPU testing. Because this is larger and slower than several nano-level YOLO variants, PWM-Net is described as computationally efficient rather than strictly nano-lightweight.

513 6.1 Limitations

514 Several limitations remain. The CQ-Fire dataset is primarily collected from a spe-
515 cific regional wildfire event, which may limit geographic diversity. Although the model
516 shows strong generalization across D-Fire and CQ-Fire, broader cross-regional valida-
517 tion would further strengthen reliability. In addition, extremely dense smoke or severe
518 occlusion may still challenge detection performance due to visual ambiguity. Opera-
519 tional deployment requires further end-to-end evaluation of camera streaming, flight
520 vibration, thermal limits, weather, altitude changes, and alarm latency.

521 The proposed architecture demonstrates that explicit frequency-domain modeling
522 combined with lightweight structural design provides a promising direction for early
523 wildfire detection in UAV-based monitoring systems.

524 7 Conclusion

525 This paper proposes PWM-Net, a computationally efficient forest fire detection
526 framework that integrates a PartialNet backbone, wavelet-based frequency-domain
527 convolution (WTConv), and a multi-spectral multi-scale attention mechanism (M2S)
528 for UAV-based early wildfire monitoring. By jointly modeling spatial, frequency, and
529 scale-domain information, the proposed method effectively enhances the detection of
530 small-scale flames and thin smoke under complex environmental conditions.

531 Experiments on public D-Fire and self-built CQ-Fire compare PWM-Net with
532 YOLO-series and Transformer-based detectors using repeated runs, strict localization
533 metrics, and resource measurements. The ablation results confirm the complementary
534 contributions of the three modules.

535 The results indicate that incorporating explicit frequency-domain feature modeling
536 and scale-aware attention mechanisms can significantly improve early fire detection
537 performance in UAV imagery. The proposed framework provides an effective solution
538 for real-time forest fire monitoring, with further gains expected from platform-specific
539 optimization and broader field validation.

540 Future work will focus on further lightweight optimization through pruning and
541 quantization, as well as expanding cross-regional datasets to enhance robustness and
542 adaptability under diverse environmental conditions.

543 Declarations

544 **Ethical approval.** Not applicable. This study does not involve human participants,
545 human biological material, identifiable personal data, or animals.

546 **Consent to participate.** Not applicable.

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Data availability. The D-Fire dataset analysed during the current study is available in the D-Fire repository, <https://github.com/gaia-solutions-on-demand/DFireDataset>. The CQ-Fire dataset generated and analysed during the current study is available from the corresponding author on reasonable request.

Code availability. The code supporting the findings of this study is available from the corresponding author on reasonable request.

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